1. What fundamental optical phenomenon governs light transmission through an optical fiber?

(A) Refraction and diffraction
(B) Total Internal Reflection (TIR)
(C) Rayleigh scattering
(D) Constructive interference

2. For total internal reflection to occur, light must travel from a:

(A) Rarer medium to a denser medium
(B) Vacuum into glass at 90 degrees
(C) Medium of low index to high index
(D) Denser medium to a rarer medium at angle > critical angle

3. The critical angle theta_c at a core-cladding boundary (core index n1, cladding index n2) is:

(A) arcsin(n2 / n1)
(B) arccos(n2 / n1)
(C) arctan(n1 / n2)
(D) arcsin(n1 / n2)

4. In all practical communication and sensing optical fibers, the core refractive index n1 must be:

(A) Equal to cladding index n2
(B) Less than cladding index n2
(C) Strictly greater than cladding index n2
(D) Zero

5. The acceptance angle theta_a of an optical fiber is defined as the:

(A) Maximum launch angle in air for which light is guided by TIR
(B) Minimum incident angle inside the core
(C) Critical angle of the core-cladding interface
(D) Brewster angle of the fiber end-face

6. Numerical Aperture (NA) of a step-index optical fiber in air (n0 = 1) is given by:

(A) n1 - n2
(B) sqrt(n1^2 + n2^2)
(C) sqrt(n1^2 - n2^2)
(D) (n1 - n2) / n1

7. The fractional refractive index difference Delta of an optical fiber is defined as:

(A) (n1 - n2) / n2
(B) (n1 - n2) / n1
(C) n1 / n2
(D) sqrt(n1 - n2)

8. The relationship between Numerical Aperture (NA) and fractional index difference Delta is:

(A) NA ~= n1 * Delta
(B) NA ~= sqrt(n1 * Delta)
(C) NA ~= 2 * n1 * Delta^2
(D) NA ~= n1 * sqrt(2 * Delta)

9. What primary material is used to fabricate low-loss telecommunication and sensing fibers?

(A) Polycrystalline quartz
(B) Optical plastic acrylic
(C) High-purity fused silica glass (SiO2)
(D) Crown glass

10. In a Step-Index (SI) fiber, the refractive index profile is:

(A) Constant throughout the core with an abrupt step at cladding
(B) Parabolic radial variation
(C) Linear decrease from center to edge
(D) Sinusoidal variation

11. In a Graded-Index (GI) fiber, the refractive index of the core:

(A) Increases radially towards the cladding
(B) Remains strictly uniform throughout
(C) Varies randomly with radius
(D) Decreases continuously from the axis towards the cladding

12. Why do graded-index fibers exhibit significantly lower modal dispersion than step-index fibers?

(A) They support only a single optical mode
(B) Rays traveling longer outer paths travel faster in lower index glass
(C) Total internal reflection is eliminated
(D) Core diameter is smaller than wavelength

13. What is the typical core diameter of a standard Single-Mode Fiber (SMF)?

(A) 50 to 62.5 micrometers
(B) 8 to 10 micrometers
(C) 100 to 200 micrometers
(D) 1.0 millimeter

14. What is the standard outer cladding diameter for almost all commercial telecom optical fibers?

(A) 125 micrometers
(B) 250 micrometers
(C) 50 micrometers
(D) 500 micrometers

15. The normalized frequency parameter (V-number) of an optical fiber is given by:

(A) V = (pi * a / lambda) * n1
(B) V = (2 * pi / lambda) * (n1 - n2)
(C) V = (2 * pi * a / lambda) * NA
(D) V = a * NA / lambda

16. What is the maximum cut-off value of V for single-mode operation in a step-index fiber?

(A) V <= 3.832
(B) V <= 1.000
(C) V <= 5.000
(D) V <= 2.405

17. For a multi-mode step-index fiber with large V, the approximate number of guided modes M is:

(A) V^2 / 4
(B) V / 2
(C) 2 * V^2
(D) V^2 / 2

18. For a parabolic graded-index fiber, the number of guided modes M is approximately:

(A) V^2 / 2
(B) V^2
(C) V^2 / 4
(D) V / 4

19. Signal attenuation in optical fibers is expressed logarithmically in units of:

(A) Decibels per kilometer (dB/km)
(B) Watts per meter (W/m)
(C) Neper per second (Np/s)
(D) Percentage transmission per meter

20. The formula for attenuation alpha (in dB/km) over fiber length L (km) is:

(A) (20 / L) * log10(P_in / P_out)
(B) (10 / L) * log10(P_in / P_out)
(C) (1 / L) * ln(P_in / P_out)
(D) (10 / L) * (P_in - P_out)

21. Rayleigh scattering loss in silica optical fibers is inversely proportional to:

(A) lambda (wavelength)
(B) lambda^4 (fourth power of wavelength)
(C) lambda^2
(D) sqrt(lambda)

22. What causes Rayleigh scattering in high-purity silica optical fibers?

(A) Air bubbles in the fiber preform
(B) Surface scratches on the cladding
(C) Core-cladding interface ovality
(D) Microscopic density and refractive index fluctuations frozen during drawing

23. The prominent attenuation peak near 1380 nm in legacy optical fibers is caused by:

(A) Hydroxyl (OH-) ion impurity absorption
(B) Transition metal ion absorption
(C) Silicon-oxygen vibrational band
(D) Rayleigh scattering

24. The minimum intrinsic attenuation window in modern silica fiber (~ 0.2 dB/km) occurs at:

(A) 1310 nm
(B) 850 nm
(C) 1550 nm (telecom C-band)
(D) 633 nm

25. Dispersion in optical fibers causes:

(A) Temporal broadening (spreading) of light pulses as they propagate
(B) Total attenuation of optical power
(C) Color changes of the laser light
(D) Depolarization into unpolarized light

26. Intermodal dispersion occurs exclusively in:

(A) Single-mode optical fibers
(B) Graded-index single-mode fibers
(C) Multi-mode optical fibers
(D) Planar photonic waveguides

27. Intramodal (chromatic) dispersion consists of two components, namely:

(A) Intermodal and Polarization dispersion
(B) Material dispersion and Waveguide dispersion
(C) Rayleigh and Raman dispersion
(D) Absorption and Scattering dispersion

28. In standard single-mode silica fibers (G.652), zero chromatic dispersion naturally occurs near:

(A) 1550 nm
(B) 850 nm
(C) 1625 nm
(D) 1310 nm

29. Macrobending loss in an optical fiber occurs when:

(A) Microscopic axial perturbations occur along the core
(B) The fiber is subjected to uniform hydrostatic pressure
(C) The fiber is bent around a radius comparable to a few centimeters
(D) The cladding is stripped off

30. What is a Fiber Bragg Grating (FBG)?

(A) A periodic modulation of core refractive index acting as a wavelength-selective mirror
(B) A prism fused to the fiber end face
(C) A metallic coating on the fiber cladding
(D) A localized microbend sensor

31. The Bragg resonance reflection condition relates Bragg wavelength lambda_B to grating pitch Lambda as:

(A) lambda_B = n_eff * Lambda
(B) lambda_B = 2 * Lambda / n_eff
(C) lambda_B = Lambda / (2 * n_eff)
(D) lambda_B = 2 * n_eff * Lambda

32. When an FBG sensor undergoes mechanical tensile strain, the reflected Bragg wavelength:

(A) Shifts toward shorter wavelengths (blue-shift)
(B) Shifts toward longer wavelengths (red-shift)
(C) Remains strictly unchanged
(D) Splits into two separate lines

33. The typical strain sensitivity of an FBG sensor operating at 1550 nm is approximately:

(A) 1.2 nanometer per microstrain
(B) 1.2 picometer per microstrain (pm / muepsilon)
(C) 0.12 picometer per microstrain
(D) 12.0 picometer per microstrain

34. The temperature sensitivity of an unbonded silica FBG at 1550 nm is approximately:

(A) 10 to 13 picometers per degree Celsius
(B) 1.0 picometer per degree Celsius
(C) 100 picometers per degree Celsius
(D) 0.1 picometer per degree Celsius

35. What technique allows dozens of FBG sensors along a single optical fiber to be interrogated independently?

(A) Spatial multiplexing only
(B) Polarization division only
(C) Wavelength Division Multiplexing (WDM)
(D) Phase inversion modulation

36. In civil infrastructure monitoring, FBG sensors embedded in concrete bridges measure:

(A) Concrete moisture pH directly
(B) Electrical conductivity of rebar
(C) Sound reverberation time
(D) Internal strain, rebar stress, crack opening, and temperature

37. Optical Time Domain Reflectometry (OTDR) measures fiber properties and locates faults by detecting:

(A) Transmitted optical power only
(B) Cavity standing wave patterns
(C) Diffraction fringe angles
(D) Rayleigh backscattered light and Fresnel reflections versus time

38. Brillouin Optical Time Domain Analysis (BOTDA) provides:

(A) Point measurements at discrete grating locations
(B) Only vibration frequency tracking
(C) Continuous distributed strain and temperature measurement over tens of kilometers
(D) Zero dispersion compensation

39. Why are optical fiber sensors inherently superior to electrical strain gauges in harsh civil environments?

(A) Immune to EMI/lightning, non-corrosive, chemically inert, and multiplexable over long spans
(B) Cheaper than copper wire
(C) They require high-voltage power lines
(D) They do not require optical interrogators

40. Which optical dopant is added to silica during chemical vapor deposition to increase core refractive index?

(A) Boron trioxide (B2O3)
(B) Germanium dioxide (GeO2)
(C) Fluorine (F)
(D) Pure nitrogen

41. An optical fiber has core refractive index n1 = 1.480 and cladding index n2 = 1.460. The critical angle theta_c is:

(A) 75.4 degrees
(B) 80.6 degrees
(C) 85.2 degrees
(D) 65.8 degrees

42. For the optical fiber in Q41, the Numerical Aperture (NA) in air (n0 = 1.0) is:

(A) 0.142
(B) 0.342
(C) 0.020
(D) 0.242

43. For the optical fiber in Q41, the acceptance angle theta_a in air is:

(A) 14.0 degrees
(B) 24.2 degrees
(C) 8.5 degrees
(D) 18.2 degrees

44. Calculate the fractional refractive index difference Delta for the fiber in Q41:

(A) 0.0270 (2.70%)
(B) 0.0068 (0.68%)
(C) 0.0135 (1.35%)
(D) 0.0200 (2.00%)

45. A step-index fiber has core radius a = 4.5 micrometers, n1 = 1.465, and n2 = 1.460. The V-number at lambda = 1.31 micrometers is:

(A) 2.61 (Multimode)
(B) 1.85 (Single mode)
(C) 3.14
(D) 2.405

46. For the fiber in Q45, what is the cut-off wavelength lambda_c for single-mode operation (V = 2.405)?

(A) 1.31 micrometers
(B) 1.55 micrometers
(C) 1.42 micrometers (1420 nm)
(D) 0.85 micrometers

47. A step-index multi-mode fiber has V = 25 at operating wavelength 850 nm. The total number of guided modes M is approximately:

(A) 625 modes
(B) 312 modes
(C) 156 modes
(D) 50 modes

48. For a parabolic graded-index fiber with the same V = 25, the number of guided modes is:

(A) 312 modes
(B) 78 modes
(C) 625 modes
(D) 156 modes

49. An optical signal of power P_in = 10 mW enters a 5.0 km optical fiber. If the output power is P_out = 2.0 mW, the attenuation coefficient alpha is:

(A) 0.70 dB/km
(B) 2.80 dB/km
(C) 1.40 dB/km
(D) 3.50 dB/km

50. A 1550 nm single-mode fiber link has an attenuation of 0.22 dB/km. What is the total optical loss across a 40 km bridge monitoring link?

(A) 8.80 dB
(B) 4.40 dB
(C) 17.6 dB
(D) 2.20 dB

51. If 1.0 mW optical power is launched into the 40 km link in Q50, the power received at the far end is:

(A) 0.500 mW
(B) 0.050 mW
(C) 0.010 mW
(D) 0.132 mW (-8.8 dBm)

52. In a 1.0 km step-index fiber with n1 = 1.50 and Delta = 0.01, the intermodal pulse broadening Delta_t_modal is approximately:

(A) 25.0 ns/km
(B) 50.0 ns/km
(C) 100 ns/km
(D) 5.0 ns/km

53. For a graded-index fiber with the identical n1 = 1.50 and Delta = 0.01, the intermodal dispersion drops to approximately:

(A) 1.25 ns/km
(B) 0.0625 ns/km (62.5 ps/km)
(C) 6.25 ns/km
(D) 12.5 ns/km

54. A silica optical fiber has a Rayleigh scattering loss of 1.5 dB/km at lambda = 850 nm. What is its Rayleigh scattering loss at 1550 nm?

(A) 0.136 dB/km
(B) 0.450 dB/km
(C) 0.820 dB/km
(D) 0.050 dB/km

55. An FBG sensor has an effective core index n_eff = 1.450 and grating period Lambda = 534.5 nm. Its Bragg reflection wavelength lambda_B is:

(A) 1310.0 nm
(B) 1064.0 nm
(C) 1550.0 nm
(D) 775.0 nm

56. If the FBG in Q55 is subjected to a tensile strain of 500 microstrain (muepsilon), what is the shift in Bragg wavelength Delta_lambda_B?

(A) +0.12 nm
(B) +1.20 nm
(C) +0.06 nm
(D) +0.60 nm (+600 pm)

57. If the temperature of the FBG in Q55 increases by 25 C without mechanical strain, the thermal wavelength shift is approximately:

(A) +50 pm
(B) +500 pm
(C) +25 pm
(D) +250 pm (+0.25 nm)

58. To isolate pure mechanical strain from temperature shifts in concrete structures, engineers install:

(A) An optical isolator
(B) A polarization controller
(C) A matched pair of FBGs (one strain-free for temperature compensation)
(D) A bandpass filter

59. A single-mode fiber has chromatic dispersion parameter D = 17 ps/(nm*km) at 1550 nm. For a laser spectral width Delta_lambda = 0.1 nm over 50 km, total pulse spreading is:

(A) 85 ps
(B) 170 ps
(C) 42.5 ps
(D) 850 ps

60. The maximum bit rate B of the fiber link in Q59 (limited by Delta_t = 85 ps, with B ~= 1 / (4 * Delta_t)) is approximately:

(A) 1.00 Gbps
(B) 2.94 Gbps
(C) 10.0 Gbps
(D) 500 Mbps

61. A plastic optical fiber (POF) has core index n1 = 1.492 and cladding index n2 = 1.402. The Numerical Aperture is:

(A) 0.310
(B) 0.510
(C) 0.710
(D) 0.090

62. For the POF in Q61, what is the acceptance cone half-angle in air?

(A) 18.1 degrees
(B) 45.2 degrees
(C) 12.5 degrees
(D) 30.7 degrees

63. An optical power meter reads an input power of 500 microwatts and an output power of 50 microwatts across a cable. The optical loss in dB is:

(A) 10.0 dB
(B) 3.0 dB
(C) 20.0 dB
(D) 6.0 dB

64. An OTDR trace shows a localized reflection spike followed by a 0.3 dB step loss at distance 3.2 km. This signature indicates:

(A) Continuous microbending along the cable
(B) Fiber end fracture
(C) A mechanical or fusion splice with slight misalignment
(D) Core diameter enlargement

65. What is the critical bend radius R_c below which macrobending loss in a single-mode fiber escalates exponentially?

(A) Approximately 2 to 3 cm
(B) 1.0 meter
(C) 0.1 mm
(D) 10 meters

66. In a polarimetric optical fiber sensor, external lateral stress on the fiber induces:

(A) Permanent core fracture
(B) Change in numerical aperture
(C) Optical birefringence (phase shift between orthogonal polarizations)
(D) Wavelength frequency doubling

67. The core radius of a single-mode fiber is a = 4.1 micrometers. At 1550 nm, its mode field diameter (MFD) is typically:

(A) Exactly 8.2 micrometers
(B) Around 9.2 to 10.4 micrometers (extends into cladding)
(C) 4.1 micrometers
(D) 125 micrometers

68. Why does mode field diameter (MFD) increase as wavelength increases in a single-mode fiber?

(A) The core physically expands with wavelength
(B) Refractive index increases with wavelength
(C) Cladding absorption forces expansion
(D) Light penetrates deeper into the cladding at longer wavelengths (V decreases)

69. In a Brillouin distributed fiber sensor, the Brillouin frequency shift nu_B in standard silica fiber at 1550 nm is approximately:

(A) 1.0 GHz
(B) 50 GHz
(C) 11 GHz
(D) 100 MHz

70. The strain coefficient of the Brillouin frequency shift in silica fiber is approximately:

(A) 0.05 MHz per microstrain (50 kHz / muepsilon)
(B) 0.5 MHz per microstrain
(C) 5.0 MHz per microstrain
(D) 0.005 MHz per microstrain

71. The temperature coefficient of the Brillouin frequency shift in silica fiber is approximately:

(A) 10.0 MHz per degree Celsius
(B) 0.1 MHz per degree Celsius
(C) 0.01 MHz per degree Celsius
(D) 1.0 MHz per degree Celsius

72. An FBG sensor interrogator uses a swept laser tuning across 1510 nm to 1590 nm (80 nm bandwidth). If each FBG requires a 4 nm operational window, what is the maximum number of multiplexed FBGs per fiber channel?

(A) 40 sensors
(B) 20 sensors
(C) 10 sensors
(D) 80 sensors

73. A concrete crack monitoring FBG with gauge length L_g = 100 mm spans an expansion joint. If a 1.2 pm wavelength shift corresponds to 1 microstrain, what wavelength shift occurs when the crack opens by 0.05 mm?

(A) +120 pm
(B) +600 pm (+0.60 nm)
(C) +1.20 nm
(D) +60 pm

74. What optical phenomenon enables distributed temperature sensing (DTS) using Raman scattering in optical fibers?

(A) Ratio of Anti-Stokes to Stokes Raman backscatter intensities
(B) Brillouin acoustic frequency shift
(C) Rayleigh phase shift
(D) Fresnel reflection amplitude

75. Why is the Anti-Stokes Raman line temperature-sensitive while the Stokes line is virtually temperature-independent?

(A) Stokes line is absorbed by silica cladding
(B) Anti-Stokes light travels at higher group velocity
(C) Anti-Stokes transitions originate from thermally populated optical phonon states
(D) Stokes light is polarized orthogonally

76. A 120-meter prestressed concrete bridge girder is monitored by an embedded FBG sensor array. During post-tensioning, the steel tendon reaches a tensile strain of 2400 microstrain while ambient temperature drops by 8 C. If the FBG strain sensitivity is 1.20 pm/muepsilon and thermal sensitivity is 10.5 pm/C, what is the net Bragg wavelength shift?

(A) +2.964 nm
(B) +2.628 nm
(C) +3.120 nm
(D) +2.796 nm

77. A fiber optic cable embedded along a 5.0 km earthen embankment dam is interrogated by BOTDA. The system achieves a spatial resolution of 1.0 m and frequency measurement precision of 0.25 MHz. What is the minimum detectable structural settlement strain?

(A) 10.0 microstrain
(B) 1.0 microstrain
(C) 25.0 microstrain
(D) 5.0 microstrain

78. A high-speed train bridge experiences dynamic live-load vibrations. An FBG interrogator sampling at 1000 Hz records a sinusoidal wavelength oscillation between 1550.12 nm and 1550.36 nm at frequency 4.5 Hz. What is the dynamic peak-to-peak strain amplitude experienced by the bridge girder?

(A) 100 microstrain
(B) 400 microstrain
(C) 200 microstrain
(D) 50 microstrain

79. A 10 km optical fiber cable is deployed inside a trans-alpine road tunnel for distributed fire detection using Raman DTS. If the fiber attenuation is 0.35 dB/km at the Stokes wavelength (1450 nm) and 0.40 dB/km at the Anti-Stokes wavelength (1350 nm), what is the differential round-trip attenuation correction over 8.0 km?

(A) 0.80 dB
(B) 0.40 dB
(C) 1.60 dB
(D) 0.20 dB

80. A concrete gravity dam undergoes curing during construction. Mass concrete heat of hydration raises the internal core temperature by 45 C over 7 days. If the embedded FBG is bonded to cured concrete with thermal expansion coefficient alpha_c = 10 x 10^-6 C^-1, what is the total wavelength shift including both concrete thermal expansion and FBG intrinsic thermo-optic effects (total effective sensitivity ~= 22 pm/C)?

(A) +0.450 nm
(B) +0.990 nm
(C) +1.980 nm
(D) +0.225 nm

81. A single-mode fiber link of length L = 80 km connects structural sensor hubs on a suspension bridge. The fiber has alpha = 0.20 dB/km, with 4 fusion splices (0.05 dB each) and 2 connector pairs (0.25 dB each). If the transmitter launches 0 dBm (1.0 mW) and receiver sensitivity is -25 dBm, what is the available optical power safety margin?

(A) 4.15 dB
(B) 8.30 dB
(C) 12.5 dB
(D) 16.6 dB

82. A Step-Index multi-mode fiber in an industrial plant monitoring network has n1 = 1.480, Delta = 0.015, and length 2.0 km. An optical pulse of FWHM 2.0 ns is launched. What is the output pulse width resulting from intermodal dispersion alone?

(A) 74 ns
(B) 296 ns
(C) 37 ns
(D) 148 ns

83. For the multi-mode fiber in Q82, if replaced by an optimized parabolic graded-index fiber with the identical parameters, what is the new intermodal pulse broadening over 2.0 km?

(A) 0.555 ns
(B) 2.78 ns
(C) 11.1 ns
(D) 0.111 ns

84. An FBG strain gauge is bonded onto a steel rebar (Young's modulus E = 200 GPa) inside a reinforced concrete column. When an axial load is applied to the column, the reflected Bragg wavelength shifts by -0.36 nm. What is the compressive stress in the rebar?

(A) 120.0 MPa
(B) 30.0 MPa
(C) 60.0 MPa (Compression)
(D) 150.0 MPa

85. A smart geotextile sheet containing integrated FBG sensors is laid beneath a railway subgrade to detect sinkhole subsidence. If an FBG experiences a curvature bend radius R = 0.50 m, what is the mechanical strain on the fiber core located 62.5 micrometers from the neutral bend axis?

(A) 125 microstrain
(B) 250 microstrain
(C) 62.5 microstrain
(D) 500 microstrain

86. An optical fiber Fabry-Perot interferometer (FPI) displacement sensor embedded in an expansion joint has an initial air cavity length d = 50 micrometers. If the expansion joint opens causing d to increase by 2.325 micrometers, how many complete interference fringes shift at lambda = 1550 nm?

(A) 1.5 fringes
(B) 6.0 fringes
(C) 3.0 fringes
(D) 4.5 fringes

87. A marine jetty pile in a coastal port is subjected to tidal corrosion. An array of 5 FBGs is inscribed with Bragg wavelengths spaced by 5.0 nm: 1530 nm, 1535 nm, 1540 nm, 1545 nm, 1550 nm. If maximum design strain is +- 1500 microstrain and temperature variation is +- 30 C, what is the maximum anticipated wavelength excursion for any individual sensor?

(A) +- 4.23 nm
(B) +- 2.115 nm
(C) +- 1.05 nm
(D) +- 5.00 nm

88. In the jetty monitoring system of Q87, is the 5.0 nm spectral spacing between adjacent FBG sensors sufficient to completely prevent spectral overlapping (crosstalk)?

(A) No, adjacent sensors will collide and overlap
(B) Only if temperature is held constant
(C) Only for tension, not compression
(D) Yes, safety margin of 0.77 nm exists between adjacent excursion bands

89. A polarization-maintaining (PM) PANDA fiber used in an optical fiber gyroscope for tunneling navigation has a beat length L_b = 2.0 mm at 1550 nm. What is the modal birefringence B = |n_x - n_y| of the fiber?

(A) 3.88 x 10^-4
(B) 1.55 x 10^-3
(C) 7.75 x 10^-4
(D) 7.75 x 10^-5

90. An OTDR pulses 10 ns duration light into a fiber cable. If the group refractive index of the core is n_g = 1.468, what is the two-point spatial event dead zone of the instrument?

(A) 1.02 meters
(B) 2.04 meters
(C) 0.51 meters
(D) 5.10 meters

91. A hollow-core photonic crystal fiber (HC-PCF) guides light through air by photonic bandgap effects. If the air core has n = 1.0003, what is the propagation speed of the optical signal compared to standard silica fiber (n = 1.45)?

(A) Same speed
(B) 30% slower
(C) Twice as fast
(D) 45% faster (v ~= 2.999 x 10^8 m/s vs 2.069 x 10^8 m/s)

92. Why are hollow-core photonic crystal fibers revolutionary for earthquake early-warning seismic sensor networks?

(A) They do not require laser sources
(B) Low-latency transmission provides critical seconds of earlier advance warning before seismic shear waves arrive
(C) They can withstand physical severance
(D) They have infinite numerical aperture

93. An optical fiber acoustic emission (AE) sensor detects high-frequency ultrasound waves (100 kHz) generated by concrete fracture. If the ultrasound wave has velocity v_p = 4000 m/s in concrete, what is the acoustic wavelength?

(A) 4.0 mm
(B) 40 mm (4.0 cm)
(C) 400 mm
(D) 0.4 mm

94. A 100 km distributed optical sensor deployed along a gas pipeline detects methane leakage cooling. If the optical pulse repetition rate is 1.0 kHz, what is the maximum unambiguous sensing range limited by round-trip Time-of-Flight (n_g = 1.468)?

(A) 102.1 km
(B) 51.1 km
(C) 204.2 km
(D) 25.5 km

95. In an FBG interrogator, a Fabry-Perot tunable filter (FPTF) has a free spectral range (FSR) of 100 nm and a finesse F = 1000. What is the spectral resolution (FWHM bandwidth) of the scanning filter?

(A) 1.0 nm
(B) 0.01 nm (10 pm)
(C) 0.10 nm (100 pm)
(D) 0.50 nm

96. A cable-stayed bridge stay-cable contains an embedded optical fiber sensor along its 250 m length. When cable tension increases from 2.0 MN to 3.5 MN, the axial elongation is 45 mm. What is the average strain recorded by the fiber?

(A) 360 microstrain
(B) 90 microstrain
(C) 450 microstrain
(D) 180 microstrain

97. For the cable in Q96, if cross-sectional steel area is A_s = 5000 mm^2 and Young's modulus E = 205 GPa, the theoretical elastic strain Delta_epsilon is:

(A) 292.6 microstrain
(B) 73.2 microstrain
(C) 219.5 microstrain
(D) 146.3 microstrain

98. A long-gauge fiber optic deformation sensor (gauge length 2.0 m) is installed on the tensile flange of a steel-concrete composite box girder. If the sensor detects an extension of 0.80 mm during proof load testing, the average bending strain is:

(A) 200 microstrain
(B) 800 microstrain
(C) 400 microstrain
(D) 100 microstrain

99. A fiber optic sensor cable embedded in asphalt highway pavement monitors wheel axle loads. When an 80 kN standard axle passes over, the induced dynamic vertical compressive strain in the asphalt layer is 150 microstrain. If the dynamic resilient modulus of asphalt is 4000 MPa, the vertical dynamic stress is:

(A) 0.60 MPa (600 kPa)
(B) 1.20 MPa
(C) 0.30 MPa
(D) 2.40 MPa

100. In fiber optic distributed acoustic sensing (DAS) based on phase-sensitive OTDR (phi-OTDR), coherent Rayleigh backscatter interference fringes monitor ground vibrations. The technique effectively transforms a standard fiber cable into an array of:

(A) Discrete temperature thermometers
(B) Thousands of continuous virtual seismic geophones spaced every 1 to 5 meters
(C) Hydrostatic pressure gauges
(D) Electric current transformers

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the optical waveguiding mechanism of total internal reflection (TIR) in optical fibers. State the conditions required for TIR and derive the expression for the critical angle.

Q2. Derive from fundamental geometric optics the expressions for the Acceptance Angle and Numerical Aperture (NA) of a step-index optical fiber.

Q3. Classify optical fibers based on: (a) refractive index profile (step-index vs graded-index), and (b) mode propagation (single-mode vs multi-mode). Compare their physical dimensions and transmission characteristics.

Q4. Define the normalized frequency parameter (V-number) of an optical fiber. Derive the single-mode cutoff condition and obtain the expressions for the number of guided modes in step-index and graded-index fibers.

Q5. Explain all loss mechanisms in optical fibers: intrinsic and extrinsic absorption, Rayleigh scattering, macrobending, and microbending losses. State how modern low-loss fibers achieve 0.2 dB/km attenuation at 1550 nm.

Q6. Derive the expression for intermodal (modal) dispersion in a step-index multi-mode optical fiber. Explain how graded-index profiling drastically mitigates pulse broadening.

Q7. Explain chromatic (intramodal) dispersion in single-mode fibers. Distinguish between material dispersion and waveguide dispersion, and explain dispersion shifting.

Q8. Explain the operating principles, mathematical formulation, and fabrication of Fiber Bragg Grating (FBG) sensors. Describe the Bragg resonance condition.

Q9. Explain the working principles of Distributed Fiber Optic Sensing based on: (a) Optical Time Domain Reflectometry (OTDR), and (b) Brillouin Optical Time Domain Analysis (BOTDA) for civil infrastructure monitoring.

Q10. Explain how optical fiber sensors are deployed for Structural Health Monitoring (SHM) of bridges, concrete dams, and tunnels. Describe installation techniques and environmental packaging.

Q11. A step-index fiber has a core of radius a = 25 micrometers with n1 = 1.485 and cladding of n2 = 1.470. At operating wavelength lambda = 1.30 micrometers, calculate: (a) critical angle, (b) numerical aperture, (c) acceptance angle in air, (d) fractional index difference, (e) V-number and

Q12. Calculate the cut-off wavelength lambda_c and core diameter for a single-mode step-index fiber with n1 = 1.468 and Delta = 0.003 to ensure single-mode propagation at all wavelengths above 1200 nm.

Q13. An optical signal of wavelength 1550 nm loses 85% of its power after traveling through a 35 km long fiber link. Calculate: (a) output power if input is 5.0 mW, (b) total loss in dB, (c) attenuation coefficient alpha in dB/km, (d) required link length for power to drop to 1% of input.

Q14. Compare the intermodal pulse broadening and maximum achievable transmission bit rate over a 10 km link for: (a) a step-index fiber with n1 = 1.50 and Delta = 0.01, and (b) an optimized parabolic graded-index fiber with the identical parameters.

Q15. An FBG sensor with lambda_B = 1550.0 nm is bonded to a structural steel girder (thermal expansion coefficient alpha_s = 12 x 10^-6 C^-1). The FBG has photoelastic coefficient p_e = 0.22 and thermo-optic coefficient xi = 6.8 x 10^-6 C^-1. When a heavy vehicle crosses, the recorded Bragg

Q16. Explain the temperature-strain cross-sensitivity problem in FBG sensors. Describe three practical optical and mechanical techniques to achieve complete temperature compensation.

Q17. A 60 km single-mode fiber link has an attenuation of 0.20 dB/km and chromatic dispersion D = 17 ps/(nm*km) at 1550 nm. It is driven by an optical transmitter with spectral linewidth 0.15 nm. Calculate: (a) total optical attenuation, (b) total chromatic dispersion pulse broadening, (c)

Q18. Explain the physical principles and architecture of Phase-Sensitive Optical Time Domain Reflectometry (phi-OTDR) used in Distributed Acoustic Sensing (DAS) for geotechnical pipeline and railway monitoring.

Q19. A mass concrete gravity dam foundation has a 200 m long embedded FBG sensor string to monitor crack development. During impoundment, an FBG indicates a localized strain jump of +1850 microstrain. If the gauge length of this sensor section is L_g = 50 mm, estimate the physical

Q20. Explain the optical construction and operation of an Extrinsic Fabry-Perot Interferometer (EFPI) fiber sensor. Derive its interference formula and explain its application in measuring microscopic expansion joint movements.

Q21. A fiber optic gyroscope (FOG) deployed in a tunnel boring machine guidance system uses a circular fiber coil of diameter D = 15 cm with 2000 turns of single-mode fiber. At wavelength lambda = 1310 nm, calculate the Sagnac phase shift Delta_phi_s when the TBM turns at an angular rate of Omega =

Q22. Explain the optical construction and spectral interrogation of a Wavelength Division Multiplexed (WDM) FBG sensor network on a highway bridge. Discuss maximum multiplexing capacity.

Q23. A submarine optical telecommunication cable deployed across a deep river estuary experiences water ingress and cable stretching. Describe how OTDR and BOTDA can pinpoint the exact fault location.

Q24. Explain the optical mechanism of Distributed Temperature Sensing (DTS) using spontaneous Raman scattering in optical fibers, and derive the temperature-dependent anti-Stokes/Stokes ratio.

Q25. Calculate the acceptance angle and numerical aperture of an optical fiber when submerged in water (n_water = 1.333), if the fiber has core index n1 = 1.480 and cladding index n2 = 1.455. Contrast with its values in air.

Q26. Discuss Polarization Mode Dispersion (PMD) in single-mode optical fibers. Explain its origin from core asymmetry and external structural stress, and state its impact on high-speed civil telemetry links.

Q27. Explain the optical characteristics and advantages of Polymer Optical Fibers (POF) compared to silica glass fibers. Discuss their niche applications in civil building networks and large-strain crack monitoring.

Q28. A multi-channel FBG interrogator records data from 12 sensors embedded in a concrete deck slab. Formulate the matrix equation for simultaneous determination of 2D strain components (epsilon_x, epsilon_y, gamma_xy) from a delta strain rosette.

Q29. An optical fiber communication link for a smart dam telemetry network has a length of 50 km and uses fiber with alpha = 0.22 dB/km. The system contains 5 fusion splices (loss 0.08 dB each) and 2 connector pairs (0.30 dB each). The laser transmitter launches -3 dBm, and the receiver

Q30. Summarize the comprehensive role of fiber optics in modern civil engineering infrastructure, highlighting structural health monitoring, smart cities, and geophysical hazard mitigation.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[B]	**Q41:**	[B]	**Q61:**	[B]	**Q81:**	[B]
Q2:	[D]	**Q22:**	[D]	**Q42:**	[D]	**Q62:**	[D]	**Q82:**	[D]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[C]	**Q24:**	[C]	**Q44:**	[C]	**Q64:**	[C]	**Q84:**	[C]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[C]	**Q26:**	[C]	**Q46:**	[C]	**Q66:**	[C]	**Q86:**	[C]
Q7:	[B]	**Q27:**	[B]	**Q47:**	[B]	**Q67:**	[B]	**Q87:**	[B]
Q8:	[D]	**Q28:**	[D]	**Q48:**	[D]	**Q68:**	[D]	**Q88:**	[D]
Q9:	[C]	**Q29:**	[C]	**Q49:**	[C]	**Q69:**	[C]	**Q89:**	[C]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[D]	**Q31:**	[D]	**Q51:**	[D]	**Q71:**	[D]	**Q91:**	[D]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[B]	**Q92:**	[B]
Q13:	[B]	**Q33:**	[B]	**Q53:**	[B]	**Q73:**	[B]	**Q93:**	[B]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[C]	**Q35:**	[C]	**Q55:**	[C]	**Q75:**	[C]	**Q95:**	[C]
Q16:	[D]	**Q36:**	[D]	**Q56:**	[D]	**Q76:**	[D]	**Q96:**	[D]
Q17:	[D]	**Q37:**	[D]	**Q57:**	[D]	**Q77:**	[D]	**Q97:**	[D]
Q18:	[C]	**Q38:**	[C]	**Q58:**	[C]	**Q78:**	[C]	**Q98:**	[C]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[B]	**Q80:**	[B]	**Q100:**	[B]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

theta_1 + phi = 90 deg => theta_1 = 90 deg - phi.
- Therefore: sin(theta_1) = sin(90 deg - phi) = cos(phi).
- Substituting into the entrance equation:
n0 * sin(theta_0) = n1 * cos(phi).

3. Imposing the Condition for Waveguiding (TIR):
- For total internal reflection at the cladding boundary, we require phi >= theta_c, where sin(theta_c) = n2 / n1.
- When phi = theta_c, the entrance launch angle reaches its maximum permissible value, termed the Acceptance Angle theta_a:
n0 * sin(theta_a) = n1 * cos(theta_c) = n1 * sqrt(1 - sin^2(theta_c)).
- Substitute sin(theta_c) = n2 / n1:
n0 * sin(theta_a) = n1 * sqrt(1 - (n2 / n1)^2) = sqrt(n1^2 - n2^2).

4. Acceptance Angle and Numerical Aperture Formula:
- Acceptance Angle in air (n0 = 1.0):
theta_a = arcsin(sqrt(n1^2 - n2^2)).
- Numerical Aperture (NA) is defined as the sine of the acceptance angle in air:
NA = sin(theta_a) = sqrt(n1^2 - n2^2) / n0.
- For air (n0 = 1):
NA = sqrt(n1^2 - n2^2) = sqrt((n1 - n2) * (n1 + n2)).
- Using fractional index difference Delta = (n1 - n2) / n1 and approximating n1 + n2 ~= 2*n1:
NA = sqrt((n1 * Delta) * (2 * n1)) = n1 * sqrt(2 * Delta).

>> **Solution to Question 3:** Classify optical fibers based on: (a) refractive index profile (step-index vs graded-index), and (b) mode propagation (single-mode vs mu

* High V-number (V > 10), supporting hundreds to thousands of distinct guided modes.
* Easy to splice and couple to inexpensive LEDs, but restricted by modal dispersion to short distances (< 2 km).

>>Solution Store Question Bank

microns with overtones at 1383 nm, 950 nm, and 725 nm. Advanced dehydration processes ('dry fibers') eliminate the 1383 nm water peak.

2. Rayleigh Scattering Loss:
- Caused by sub-microscopic random thermal density and dopant concentration fluctuations frozen into the silica network during glass solidification from melt (~ 2000 C).
- Attenuation follows Rayleigh's law: alpha_R proportional to 1 / lambda^4.
- At 850 nm: alpha_R ~= 1.5 dB/km; at 1310 nm: alpha_R ~= 0.35 dB/km; at 1550 nm: alpha_R ~= 0.15 dB/km.

3. Radiative Bending Losses:
- (a) Macrobending Loss: Occurs when a fiber is bent around a curved structural radius comparable to centimeters. Evanescent field in cladding must travel faster than c/n2 to keep phase with core wavefront, radiating power outward into the buffer coating.
- (b) Microbending Loss: Microscopic localized axial perturbations (micron scale) along the core caused by lateral pressure, uneven cable jacketing, or temperature contraction during concrete setting, coupling guided modes into unguided radiating cladding modes.

4. Achieving 0.2 dB/km at 1550 nm:
- At 1550 nm, Rayleigh scattering has declined to ~ 0.15 dB/km while infrared silica absorption has just begun to rise (~ 0.03 dB/km).
- In pure silica core fibers with fluorine-doped cladding (eliminating germanium scattering centers), total attenuation reaches an astounding theoretical minimum of 0.15 to 0.18 dB/km.

>>**Solution to Question 6:** Derive the expression for intermodal (modal) dispersion in a step-index multi-mode optical fiber. Explain how graded-index profiling dra

- Velocity of light is v(r) = c / n(r); rays swinging further outward travel at substantially higher speeds through lower-index glass.
- The extra path length of outer swinging rays is almost perfectly compensated by their higher average speed.
- Theoretical intermodal pulse broadening reduces to:
Delta_t_GI ~= (L * n1 * Delta^2) / (8 * c).
- Because Delta is small (~ 0.01), Delta^2 ~= 10^-4, which reduces modal dispersion by a factor of ~ 1000 compared to step-index fibers.

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ultraviolet (UV) laser light (lambda ~ 244 nm or 193 nm).
- Uses two-beam holographic interferometry or a phase mask.
- Permanent UV-induced defect centers cause a permanent periodic sinusoidal modulation of the core refractive index (Delta_n ~ 10^-5 to 10^-3) with spatial pitch Lambda (~ 500 nm).

2. The Bragg Resonance Condition:
- When broadband light is launched down the fiber core, each individual index perturbation reflects a microscopic fraction of light.
- Constructive interference between successive reflected wavelets occurs when the round-trip phase difference equals an integer multiple of 2*pi:
2 * (effective path) = m * lambda_B => 2 * (n_eff * Lambda) = lambda_B (for first order m = 1).
- The Bragg Wavelength lambda_B is strictly governed by:
lambda_B = 2 * n_eff * Lambda,
where n_eff is the effective mode index and Lambda is the grating period.
- All other wavelengths pass through the grating unattenuated, while a very narrow spectral notch (~ 0.2 nm FWHM) centered at lambda_B is reflected.

3. Sensing Mechanism:
- Any external physical disturbance that alters grating pitch Lambda or refractive index n_eff directly shifts the reflected Bragg wavelength:
Delta_lambda_B = 2 * (Delta_n_eff * Lambda + n_eff * Delta_Lambda).
- Shift due to Mechanical Strain (epsilon): Physical stretching expands pitch (Delta_Lambda = Lambda * epsilon) and alters index via photoelastic effect:
(Delta_lambda_B)_strain = lambda_B * (1 - p_e) * epsilon (~ 1.2 pm / microstrain at 1550 nm).
- Shift due to Temperature (Delta_T): Thermal expansion expands pitch (alpha_th) and thermo-optic effect alters index (xi = (1/n)*dn/dT):
(Delta_lambda_B)_temp = lambda_B * (alpha_th + xi) * Delta_T (~ 10 pm / C at 1550 nm).

>>**Solution to Question 9:** Explain the working principles of Distributed Fiber Optic Sensing based on: (a) Optical Time Domain Reflectometry (OTDR), and (b) Bri

- Brillouin Frequency Shift nu_B:
nu_B = 2 * n * V_a / lambda,
where V_a is acoustic sound velocity in silica (~ 5960 m/s).
- The acoustic velocity and index vary linearly with mechanical strain (epsilon) and temperature (T):
Delta_nu_B = C_epsilon * Delta_epsilon + C_T * Delta_T,
where C_epsilon ~= 0.05 MHz / microstrain and C_T ~= 1.0 MHz / C.
- By sweeping probe frequency and recording gain vs arrival time, BOTDA provides complete strain and temperature profiles over spans > 50 km with 1-meter spatial resolution.

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2. (a) Critical Angle theta_c:
- sin(theta_c) = n2 / n1 = 1.470 / 1.485 = 0.989899.
- theta_c = arcsin(0.989899) = 81.85 degrees.

3. (b) Numerical Aperture NA:
- NA = sqrt(n1^2 - n2^2) = sqrt((1.485)^2 - (1.470)^2) = sqrt(2.205225 - 2.160900) = sqrt(0.044325) = 0.2105.

4. (c) Acceptance Angle in Air theta_a:
- sin(theta_a) = NA = 0.2105 => theta_a = arcsin(0.2105) = 12.15 degrees.
- Total acceptance cone angle: 2 * theta_a = 24.3 degrees.

5. (d) Fractional Refractive Index Difference Delta:
- Delta = (n1 - n2) / n1 = (1.485 - 1.470) / 1.485 = 0.015 / 1.485 = 0.0101 (1.01%).

6. (e) V-Number and Number of Guided Modes M:
- V = (2 * pi * a / lambda) * NA = [(2 * pi * 25 x 10^-6) / (1.30 x 10^-6)] * 0.2105 = (157.08 / 1.30) * 0.2105 = 120.83 * 0.2105 = 25.44.
- Since V = 25.44 >> 2.405, this is a multi-mode fiber.
- Number of guided modes: M ~= V^2 / 2 = (25.44)^2 / 2 = 647.2 / 2 ~= 323 modes.

>>Solution Step Question 6.12 Calculate the cut-off wavelength lambda_c for a step-index fiber with n1 = 1.468 and Delta = 0.003 to ensure single-mode operation at 1550 nm. Also, determine the mode field diameter using Marcuse's formula if the fiber has a core radius of 4.5 μm. Solution: Step 1: Given Data Core refractive index n1 = 1.468 Relative refractive index difference Delta = 0.003 Core radius a = 4.5 μm = 4.5 × 10^-6 m Operating wavelength lambda = 1550 nm = 1.55 × 10^-6 m Step 2: Calculate the Cut-off Wavelength (lambda_c) For single-mode operation, the normalized frequency V must be less than or equal to 2.405. The cut-off wavelength is the wavelength at which V = 2.405. The formula for V is: V = (2 * pi * a / lambda) * n1 * sqrt(2 * Delta) Rearranging to find lambda_c: lambda_c = (2 * pi * a * n1 * sqrt(2 * Delta)) / 2.405 Substitute the values: lambda_c = (2 * 3.14159 * 4.5 × 10^-6 * 1.468 * sqrt(2 * 0.003)) / 2.405 lambda_c = (2 * 3.14159 * 4.5 × 10^-6 * 1.468 * sqrt(0.006)) / 2.405 lambda_c = (2 * 3.14159 * 4.5 × 10^-6 * 1.468 * 0.07746) / 2.405 lambda_c = (3.2004 × 10^-6) / 2.405 lambda_c = 1.331 × 10^-6 m = 1331 nm Step 3: Calculate the Mode Field Diameter (MFD) Using Marcuse's formula for the mode field diameter (2w) at the operating wavelength: 2w = 2a * (0.65 + 1.619 / V^1.5 + 2.879 / V^6) First, calculate V at the operating wavelength: V = (2 * 3.14159 * 4.5 × 10^-6 / 1.55 × 10^-6) * 1.468 * sqrt(2 * 0.003) V = (18.2404) * 1.468 * 0.07746 V = 2.076 Now, calculate the MFD: 2w = 2 * 4.5 × 10^-6 * (0.65 + 1.619 / (2.076)^1.5 + 2.879 / (2.076)^6) 2w = 9 × 10^-6 * (0.65 + 1.619 / 2.991 + 2.879 / 73.15) 2w = 9 × 10^-6 * (0.65 + 0.541 + 0.039) 2w = 9 × 10^-6 * 1.230 2w = 11.07 × 10^-6 m = 11.07 μm Step 4: Conclusion The cut-off wavelength is 1331 nm. Since the operating wavelength (1550 nm) is greater than the cut-off wavelength, the fiber will operate in single-mode. The mode field diameter is approximately 11.07 μm, which is slightly larger than the core diameter, indicating that about 85% of the power is confined within the core.

1. Given Data:
- Input power P_in = 5.0 mW.
- Power lost = 85%, therefore power remaining = 15% = 0.15.
- Link length L = 35 km.

2. (a) Output Power P_out:
- P_out = 0.15 * P_in = 0.15 * 5.0 mW = 0.75 mW.

3. (b) Total Optical Loss in dB:
- Loss (dB) = 10 * log10(P_in / P_out) = 10 * log10(1 / 0.15) = 10 * log10(6.6667) = 10 * 0.8239 = 8.24 dB.

4. (c) Attenuation Coefficient alpha in dB/km:
- alpha = Total Loss / L = 8.239 dB / 35 km = 0.235 dB/km.

5. (d) Link Length L_1% for Power to Drop to 1% (P_out / P_in = 0.01):
- Total loss required for 1% transmission:
Loss_1% = 10 * log10(1 / 0.01) = 10 * log10(100) = 20.0 dB.
- L_1% = Loss_1% / alpha = 20.0 dB / (0.2354 dB/km) = 84.96 kilometers.

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1. Governing Strain-Wavelength Relation:
- For pure mechanical strain at constant temperature (Delta_T = 0):
Delta_lambda_B / lambda_B = (1 - p_e) * epsilon,
where p_e is the effective photoelastic coefficient and epsilon is mechanical strain.

2. (a) Calculation of Mechanical Strain epsilon:
- Given: lambda_B = 1550.0 nm; Delta_lambda_B = +0.380 nm; p_e = 0.22.
- (1 - p_e) = 1 - 0.22 = 0.78.
- epsilon = Delta_lambda_B / [lambda_B * (1 - p_e)]
- epsilon = 0.380 / [1550.0 * 0.78] = 0.380 / 1209.0 = 3.143 x 10^-4 = 314.3 microstrain (muepsilon).

3. (b) Calculation of Tensile Stress sigma_steel:
- By Hooke's Law: sigma = E * epsilon.
- E = 205 GPa = 205 x 10^9 N/m^2.
- sigma = (205 x 10^9 N/m^2) * (3.143 x 10^-4) = 6.443 x 10^7 N/m^2 = 64.43 MPa.

4. Structural Assessment:
- The measured tensile stress of 64.4 MPa is well below the yield strength of structural steel (f_y = 250 - 355 MPa), confirming that the bridge girder remains comfortably within its linear elastic design capacity.

>> Solution to Question 16 Explain the temperature-strain cross-sensitivity problem in FBG sensors. Describe three practical optical and mechanical techniques to

- Inscribing FBGs in polarization-maintaining (PM) fiber generates two reflection peaks with different strain and temperature sensitivities, solving two simultaneous algebraic equations.

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and high-pressure fluid leaks.
- Perimeter Security: Detects human intrusion along airport and dam boundary fences.

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3. Expansion Joint Application:
- When an expansion joint or structural crack opens by displacement Delta_d, the cavity expands to d + Delta_d.
- By tracking spectral interference fringe shifts using white-light interferometry, absolute displacement Delta_d is measured with sub-nanometer resolution.
- EFPI sensors are completely immune to thermal expansion of the lead-in cable and exhibit virtually zero temperature cross-sensitivity.

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2. Spectral Interrogation and Demultiplexing:
- When broadband light pulses down the fiber, each FBG selectively reflects only its designated spectral line back to the interrogator.
- The return spectrum consists of N distinct reflection peaks in the wavelength domain.
- By tracking peak centers independently via centroid or Gaussian fitting algorithms, strain and temperature at all N locations are resolved simultaneously from a single fiber strand.

3. Maximum Multiplexing Capacity Analysis:
- Available source spectral bandwidth: Delta_lambda_source = 80 nm (e.g., 1520 to 1600 nm).
- For each sensor, allocate operational spectral bandwidth: Delta_lambda_alloc = Delta_lambda_strain + Delta_lambda_temp + Delta_lambda_guard.
- Maximum bridge strain: +- 1500 muepsilon => Delta_lambda_strain = 2 * (1500 * 1.2 pm) = 3.6 nm.
- Maximum temperature span: +- 25 C => Delta_lambda_temp = 2 * (25 * 10 pm) = 0.5 nm.
- Guard band (FWHM filter clearance): 0.9 nm.
- Total allocation per sensor: Delta_lambda_alloc = 3.6 + 0.5 + 0.9 = 5.0 nm.
- Maximum capacity per channel: N_max = 80 nm / 5.0 nm = 16 sensors per fiber.
- Using an optical 8-channel switch, a single interrogator monitors 16 * 8 = 128 structural points.

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molecular vibrational modes (optical phonons) of the Si-O-Si glass lattice.
- Raman scattering produces two inelastically shifted sidebands:
* Stokes Line (lambda_S > lambda_0): Photon surrenders energy to create an optical phonon; frequency down-shifted to nu_0 - Delta_nu_vib (Delta_nu_vib ~= 13.2 THz, ~ 440 cm^-1).
* Anti-Stokes Line (lambda_AS < lambda_0): Photon absorbs energy from an existing thermal optical phonon; frequency up-shifted to nu_0 + Delta_nu_vib.

2. Temperature Dependence Derivation:
- The probability of an optical phonon existing in the glass lattice at thermal equilibrium is governed by Bose-Einstein statistics:
n_phonon = 1 / [exp(h * Delta_nu_vib / (k_B * T)) - 1].
- (a) Stokes Intensity: Involves phonon creation (stimulated + spontaneous emission proportional to n_phonon + 1):
I_Stokes proportional to exp(h*Delta_nu_vib / (k_B*T)) / [exp(h*Delta_nu_vib / (k_B*T)) - 1] ~= constant (very weak T dependence).
- (b) Anti-Stokes Intensity: Involves phonon absorption (proportional to n_phonon alone):
I_Anti-Stokes proportional to 1 / [exp(h * Delta_nu_vib / (k_B * T)) - 1] (highly sensitive to T!).
- (c) Ratio of Intensities:
R(T) = I_Anti-Stokes / I_Stokes = (lambda_S / lambda_AS)^4 * exp(- h * Delta_nu_vib / (k_B * T)).

3. Civil Applications:
- By taking the ratio R(T), laser power fluctuations and fiber bending losses cancel out, providing absolute temperature measurements along road tunnels (fire detection), dams (leakage tracing), and power cables.

>> Solution to Question 25: Calculate the acceptance angle and numerical aperture of an optical fiber when submerged in water (n_water = 1.333), if the fiber has

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3. Limitation:
- High attenuation: Typically 100 to 150 dB/km at 650 nm, restricting transmission to short links (< 100 m).

4. Civil Engineering Applications:
- Large-Strain Concrete Crack and Fault Monitoring: Silica fibers snap when concrete cracks open beyond 0.2 mm. POF fibers stretch across seismic fault lines and bridge expansion joints, measuring massive plastic displacements up to 10-50 mm without breaking.
- Smart In-Home Automation: Rugged high-bandwidth optical wiring inside residential drywall and flooring immune to sharp bends.

>> Solution to Question 28: A multi-channel FBG interrogator records data from 12 sensors embedded in a concrete deck slab. Formulate the matrix equation for s

[Problem Statement]: An optical fiber communication link for a smart dam telemetry network has a length of 50 km and uses fiber with alpha = 0.2
[Detailed Step-by-Step Answer]:
1. Transmitter and Receiver Parameters:
- Launched optical power: P_tx = -3.0 dBm (0.50 mW).
- Receiver sensitivity: P_rx = -28.0 dBm (1.58 microwatts).
- Total allowable link loss: Loss_allowable = P_tx - P_rx = -3.0 - (-28.0) = 25.0 dB.

2. Calculation of Channel Losses:
- (a) Fiber Attenuation Loss:
Loss_fiber = alpha * L = (0.22 dB/km) * 50 km = 11.00 dB.
- (b) Splice Losses (5 fusion splices):
Loss_splices = 5 * 0.08 dB = 0.40 dB.
- (c) Connector Losses (2 connector pairs):
Loss_connectors = 2 * 0.30 dB = 0.60 dB.
- Total Channel Loss:
Loss_total = 11.00 + 0.40 + 0.60 = 12.00 dB.

3. Calculation of Optical Power Safety Margin:
- Power Margin = Loss_allowable - Loss_total = 25.00 dB - 12.00 dB = 13.00 dB.

4. Engineering Assessment:
- The available power safety margin is 13.0 dB, which substantially exceeds the required 6.0 dB industry standard margin.
- The link will operate with extraordinary reliability over decades, easily absorbing fiber aging and future repair splices.

>>**Solution to Question 30:** [Problem Statement]: Summarize the comprehensive role of fiber optics in modern civil engineering infrastructure, highlighting structural health monitoring, sm

- The immunity of optical fibers to lightning, corrosion, and electromagnetic interference makes photonics the premier technology ensuring long-term resilience, lifecycle safety, and digital transformation of 21st-century civil infrastructure.
